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Manzil: Intelligent Traffic & Emergency Routing

Deliverable 2 Report

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Muhammad Ali Mahoon	23L-0531
Muzammil Ahmed Bhatti	23L-0688
Mohammad Ibrahim Salman	23L-0723

Supervisor: Miss Umm-e-Ammarah

Department of Computer Science

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Date: _____

Muhammad Ali Mahoon:

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Muzammil Ahmed Bhatti:

_____

Mohammad Ibrahim Salman:

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Authors' Declaration

We declare that the work presented in this Deliverable 2 report is our own project work and has not been submitted previously, in whole or in part, to another institution for an academic award. This document describes a proposed system and planned evaluation; it does not present unperformed experiments as completed results.

Abstract

Manzil is a simulation-based research and development prototype for adaptive urban traffic control and emergency routing. It combines video-based vehicle detection and tracking, traffic-state estimation, comparative signal-control policies, congestion-aware ambulance routing, and a moving green corridor within a SUMO test environment. A React dashboard and FastAPI services are proposed to expose system state and experimental records. This report specifies the requirements, research method, architecture, and controlled evaluation protocol; implementation and measured results remain subsequent project work.

Executive Summary

Urban road demand changes over time and between approaches, whereas a fixed timing plan cannot respond to every short-term imbalance. Emergency vehicles face a related but stricter problem: a geographically short route can still be slow when queues, signal delay, or an incident obstruct the path. Manzil studies these two problems in one controlled prototype. The system is intended to compare fixed-time, rule-based adaptive, and reinforcement-learning signal control, then determine the contribution of dynamic routing and a moving green corridor to ambulance travel while measuring the disruption imposed on ordinary traffic.

The project deliberately separates two evidence paths. Recorded traffic video is used to study whether a detector and multi-object tracker can produce observable counts, speeds, and queue proxies. SUMO and TraCI provide exact, repeatable network state for controller and routing experiments. These paths share a versioned traffic-state contract, but this report does not claim that arbitrary camera footage reconstructs a complete road network. Camera calibration, lane mapping, observation freshness, and uncertainty are explicit constraints.

The proposed system uses a React dashboard, FastAPI application services, a background perception worker, a SUMO/TraCI adapter, controller and routing services, and persistent experimental records. PostgreSQL is a proposed persistence choice rather than a confirmed implementation fact. A selected, manageable urban network is to be imported from OpenStreetMap and cleaned in SUMO NetEdit, with a synthetic grid retained as a fallback. The core emergency case contains one active ambulance. Dynamic edge costs combine estimated travel time, queue and signal delay without treating distance alone as the objective. Priority is activated only for the next relevant intersections, uses safe clearance transitions, detects passage, and then restores normal operation.

Evaluation is planned as matched, repeated simulation runs under light, medium, and heavy demand. Normal traffic experiments compare the three controllers using travel time, waiting time, queue length, throughput, and starvation checks. Emergency experiments compare an unassisted reference, congestion-aware routing, and routing with corridor control using ambulance travel time and stops together with normal-traffic delay and recovery time. Training and tuning scenarios will be separated from held-out evaluation scenarios, and all configurations, random seeds, software versions, and failures will be retained. No performance values are reported at this Deliverable 2 stage.

This report covers the project vision, a critical review of relevant research, software requirements, the proposed research method, and high- and low-level design. Its main contribution at this stage is a traceable and reproducible specification for investigating an integrated traffic-control and emergency-

priority approach without overstating implementation maturity or real-world deployability.

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Chapter 1 Introduction

Urban traffic control is a networked decision problem. Demand can change by approach, queue spillback at one junction can obstruct another, and a timing plan that is adequate in one period may waste capacity in another. Emergency travel adds a stricter timing objective. An ambulance does not benefit merely from a short geometric path if that path contains slow links, long queues, or signals that repeatedly stop it. These observations motivate Manzil, a research and development prototype that studies traffic perception, adaptive signal control, congestion-aware emergency routing, and coordinated signal priority in one simulation-based environment.

Manzil is not presented as a deployable municipal controller. It is a controlled experimental platform based on the microscopic traffic simulator SUMO and its TraCI interface [1, 2]. The platform is intended to make policies repeatable, observable, and comparable before any real-world actuation is considered. Recorded video supplies a second, deliberately bounded input mode for studying whether vehicle detection and tracking can produce useful traffic indicators. The simulator remains the authoritative source of network state in routing and signal-control experiments.

1.1 Purpose of this Document

This Deliverable 2 report translates the approved proposal into a traceable research and software design. It defines the questions to be investigated, reviews the evidence behind major design choices, specifies externally visible requirements, and describes the methodology and architecture needed to conduct fair experiments. It also records the distinction between confirmed project scope, proposed technical choices, and future implementation work.

The report does not claim experimental improvements, real-time performance, model accuracy, or deployment readiness. At the proposal stage the problem definition, major functions, and conceptual architecture were established; dataset preparation, implementation, tuning, and evaluation were still planned work. The methodology therefore defines how later claims must be produced rather than anticipating their outcome.

1.2 Intended Audience

The primary audience is the FAST–NUCES FYP supervisor and evaluation panel. They require enough detail to judge research relevance, software scope, feasibility, and the validity of the proposed evaluation. The report is also written for the student implementation team, who should be able to derive interfaces, experiment configurations, and acceptance checks from it. A secondary audience consists of traffic-

domain and emergency-service stakeholders who may assess the assumptions and limitations, although those stakeholders are not treated as current project clients.

1.3 Project Status and Evidence Boundaries

Table 1.1 prevents proposed components from being read as completed ones. A choice is described as "proposed" where the repository contains a design decision but no implementation artifact or measured evidence. This convention is used throughout the report.

Table 1.1: Project status at Deliverable 2

Area	Confirmed evidence	Not yet evidenced
Problem and scope	Approved proposal, objectives, exclusions, and simulation basis	Field deployment or agency integration
Architecture	Conceptual React/FastAPI/SUMO and perception decomposition	Deployed services and measured capacity
Traffic control	Fixed, rule-adaptive, and RL comparison plan	Final RL policy and comparative results
Emergency operation	Dynamic route and moving-corridor design	Validated timing parameters and achieved savings
Perception	YOLO-based detector and tracker direction	Final checkpoint, annotated split, and measured accuracy
Data persistence	Experiment records and candidate entities	Confirmed database engine and schema migrations

1.4 Definitions, Acronyms, and Abbreviations

Table 1.2: Key terminology

Term	Definition in this report
A*	A shortest-path search that ranks candidates using accumulated cost plus an admissible lower-bound heuristic [3].
API	Application Programming Interface.
CV	Computer vision; here, processing traffic video into observable vehicle-level and aggregate indicators.

Term	Definition in this report
Green corridor	A sequence of bounded, moving signal-priority interventions ahead of an ambulance; not permanent green on the whole route.
MOT	Multi-object tracking, which associates detections into identities across frames.
Queue proxy	An operational estimate based on tracked, slow or stationary vehicles in a defined lane region; not a direct physical queue measurement unless calibrated.
RL	Reinforcement learning, in which a policy is optimized using state, actions, and cumulative reward.
Signal preemption	Interruption or modification of normal signal operation to serve an emergency request, including transition and recovery.
SUMO	Simulation of Urban MObility, a microscopic road-traffic simulator.
TraCI	Traffic Control Interface, used by an external process to read and change a running SUMO simulation.
Traffic-state contract	Versioned representation of approach observations, signal state, source mode, timestamps, units, confidence, and validity.
Throughput	Vehicles completing a defined movement or leaving a measured network region per unit time.
Waiting time	Time accumulated under the explicitly configured low-speed threshold; the exact SUMO definition and threshold will be recorded with each experiment.

1.5 Report Organization

Chapter 2 defines the problem, objectives, boundaries, stakeholders, constraints, and research questions. Chapter 3 critically reviews perception, traffic-control, routing, and emergency-priority research. Chapter 4 specifies functional and quality requirements, use cases, wireframes, data, risk, and traceability. Chapter 5 defines the proposed pipelines, algorithms, experimental protocol, metrics, and validity controls. Chapter 6 presents the component architecture, domain model, interactions, corridor lifecycle, contracts, concurrency model, and engineering policies.

1.6 Conclusion

Manzil is framed as a reproducible investigation of interacting traffic decisions rather than as a collection of unrelated AI features. The remainder of the report maintains a strict boundary between planned methods and completed evidence, which is essential for evaluating Deliverable 2 on the quality of its design.

Chapter 2 Project Vision

This chapter converts the broad motivation into a bounded project vision. It identifies the decisions Manzil will make, the evidence needed to judge those decisions, and the conditions under which conclusions will remain limited to simulation and selected video.

2.1 Problem Domain Overview

Traffic-signal control allocates time among incompatible movements. Fixed plans are understandable and easy to reproduce, but cannot react to short-term imbalance. Adaptive control can use measured demand, yet an apparently efficient policy can be unfair if a lightly loaded approach waits indefinitely. Network coordination complicates the problem because an action at one intersection changes arrivals at downstream intersections.

Emergency routing operates on the same network but has a different objective. A route is useful only when its costs reflect current travel time and intersection delay. Signal priority can reduce stops, but careless preemption can create unsafe phase changes or long recovery queues. The project therefore treats route selection, priority activation, passage detection, and recovery as one lifecycle while retaining separate measurements for the ambulance and ordinary traffic.

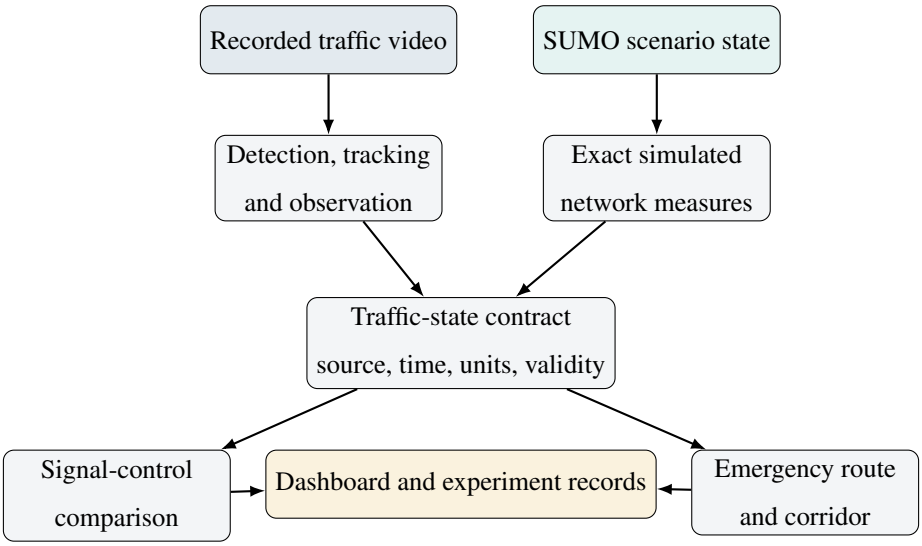


Figure 2.1: Manzil project context and the separation of evidence paths

Figure 2.1 shows the central scope decision. Video observations may be evaluated against annotations, while SUMO state drives repeatable control experiments. Both can populate a common contract, but a measurement from one source is never silently treated as ground truth from the other.

2.2 Problem Statement

The project addresses the absence, within the selected prototype setting, of a reproducible way to compare traffic-signal strategies and to quantify the combined effect of congestion-aware ambulance routing and coordinated signal priority. The problem is to design a system that observes or simulates traffic state, makes bounded control and routing decisions, records their causes and outcomes, and supports fair comparison without claiming that simulation results transfer directly to a deployed city.

2.3 Problem Elaboration

Five linked sub-problems must be controlled. First, detections must be associated over time so that frame-level boxes can become counts, trajectories, and lane-region observations. Second, normal signal control requires fixed and adaptive baselines before RL is meaningful. Third, route costs must use consistent units and must not count congestion twice. Fourth, the corridor must anticipate an ambulance without holding unnecessary intersections green. Fifth, all comparisons must use matched demand, seeds, and configurations.

An error can cross module boundaries. A stale observation may alter an edge cost; a route change may invalidate planned signal priority; a prolonged priority phase may improve ambulance time while worsening ordinary traffic; and an uncontrolled simulation clock may allow different services to act on different steps. The vision therefore includes explicit freshness, ownership, fallback, and audit requirements.

2.4 Research Questions

RQ1 With declared video sources and annotations, how accurately and consistently can the proposed detector-tracker pipeline estimate observable vehicle counts and queue-related indicators?

RQ2 Under matched SUMO demand scenarios, how do fixed-time, rule-based adaptive, and a selected RL controller differ in waiting time, queue length, travel time, throughput, and approach fairness?

RQ3 Holding signal control constant, what change in ambulance travel time, stops, and route stability is attributable to dynamic congestion-aware routing?

RQ4 Holding routing logic constant, what additional emergency benefit and ordinary-traffic disruption result from the moving green corridor?

RQ5 How sensitive are the conclusions to demand level, incident placement, random seed, and observation uncertainty?

These questions avoid a single vague claim of "better traffic." Each comparison names a controlled factor and paired outcome measures.

2.5 Goals and Objectives

The goal is to develop and evaluate a simulation-based intelligent traffic-management prototype that supports normal traffic control and a single emergency mission. The measurable objectives are:

O1. Define and implement a versioned traffic-state representation shared by perception, simulation, control, routing, and visualization.

O2. Evaluate a pretrained vehicle detector with multi-object tracking on held-out, source-separated traffic-video segments.

O3. Construct a manageable multi-intersection SUMO network and repeatable demand scenarios.

O4. Implement fixed-time and reproducible rule-adaptive controllers, followed by one feasible RL controller.

O5. Compute valid ambulance routes using time-based dynamic edge costs and bounded rerouting.

O6. Coordinate safe, moving signal priority and restore normal control after passage or cancellation.

O7. Provide a React dashboard and FastAPI service boundary for scenario control, monitoring, and experiment comparison.

O8. Record configurations and outcomes so each reported run can be reproduced and audited.

2.6 Project Scope

Table 2.1: Scope boundaries

In scope	Outside the current scope
Recorded-video vehicle detection, tracking, counts, and queue proxies	Live government CCTV integration or citywide camera calibration
Selected multi-intersection SUMO network and synthetic demand	All of Lahore or safety-certified field deployment
Fixed, rule-adaptive, and one RL controller	Replacing real traffic-controller firmware
Single active ambulance with dynamic routing and moving priority	Multiple competing emergency fleets as a core case
Simulated closure or incident as a routing trigger	Visual accident classification or traffic forecasting
React dashboard, FastAPI services, and experiment records	Native mobile application, enforcement, facial or number-plate recognition
OpenStreetMap import with manual cleanup where feasible	A guarantee that map attributes are complete or current

2.7 Sustainable Development Goals

Manzil aligns principally with United Nations Sustainable Development Goal 11, which concerns inclusive, safe, resilient, and sustainable cities [4]. The project investigates more efficient use of existing intersection capacity and more timely emergency movement, but it does not claim city-level impact from a simulator. A secondary connection exists with health outcomes because ambulance delay can matter, although the project neither measures clinical outcomes nor models medical priority.

2.8 Constraints and Assumptions

The main constraints are a semester-scale implementation window, limited access to local calibrated traffic data, camera occlusion and perspective, the simulation-to-reality gap, and finite training hardware. The selected network must therefore remain small enough for repeated trials. A real urban extract is preferred, but a controlled grid is an acceptable fallback if map signal logic is incomplete.

The core experiments assume one active ambulance, valid origin and destination edges, controlled traffic

generation, and no direct actuation of public infrastructure. Video-derived queue length is an observable proxy unless the scene is calibrated. PostgreSQL, the precise tracker, the RL algorithm, and computing hardware remain proposed choices until implementation evidence confirms them.

2.9 Business and Practical Opportunity

The prototype's practical value is as a decision-support and experimentation environment. It can demonstrate how policies react to repeatable congestion and emergency scenarios, help compare controller designs, and expose the cost imposed on non-priority traffic. Any commercial or public-sector opportunity would require validated local data, authority approval, hardware integration, cybersecurity assessment, safety engineering, and field trials. No partnership, deployment commitment, or revenue expectation is assumed.

2.10 Stakeholders and User Characteristics

Table 2.2: Stakeholder summary

Stakeholder	Interest	Expected interaction
Student research team	Correct implementation and defensible experiments	Configure, run, diagnose, export, and compare experiments
Supervisor and evaluators	Scope, rigor, traceability, and evidence quality	Review report, configurations, demonstrations, and results
Prototype operator	Clear scenario and emergency controls	Use the dashboard; inspect warnings and system state
Prospective traffic authority	Safety, reliability, network impact, auditability	Future consultation only; no current operational role
Prospective emergency service	Travel-time benefit and predictable priority behavior	Future requirements input only
Road users	Fairness and limited disruption	Indirect beneficiaries or affected parties in simulated evaluation

2.11 Success Criteria

Deliverable success is not defined by a visually impressive ambulance alone. The research prototype should execute matched scenarios without hidden configuration changes, record all interventions, reject

invalid routes and stale inputs, and report both benefits and side effects. A conclusion is considered supported only when it is tied to a named scenario set, repeated seeds, variability, and an unchanged measurement definition.

2.12 Conclusion

The vision is deliberately narrower than real traffic management: a controlled, auditable prototype with one core emergency case and explicit evidence boundaries. These boundaries shape the literature review and prevent later design choices from silently expanding the approved scope.

Chapter 3 Literature Review and Related Work

This chapter examines the research needed to justify Manzil's design rather than treating the use of familiar tools as a contribution. The review is organized around five decisions: how traffic is observed, how normal signals are controlled, how learned policies are evaluated, how emergency routes are selected, and how signal priority is activated and recovered. Reported results are interpreted within their original datasets and environments; values from incompatible benchmarks are not compared as though they were a common test.

3.1 Review Method and Terminology

The approved proposal provided an initial bibliography. Each retained study was checked against a publisher page, proceedings record, DOI, or official open-access paper. Studies were selected for direct relevance to detection and tracking, traffic-signal control, shortest-path routing, emergency routing, preemption, or simulation. Official software documentation is used for interfaces but is not counted among the research studies.

The review distinguishes *detection* from *tracking*. Detection locates and classifies objects in a frame; tracking associates those observations through time. It also distinguishes *priority* from *preemption*. Priority may adjust normal timing within constraints, whereas preemption interrupts normal operation for an emergency request. Manzil uses "corridor" for a coordinated sequence of such bounded interventions.

3.2 Vision-Based Traffic Observation

3.2.1 One-Stage Object Detection

Redmon et al. formulated YOLO as a single network that predicts bounding boxes and class probabilities from an image in one evaluation [5]. The work established the feasibility of real-time one-stage detection and reported strong speed, while also identifying localization errors as a relative weakness. For Manzil, this supports a pretrained one-stage detector as a feasible starting point, not an assumption that all vehicle classes and viewpoints will be accurate. A detector checkpoint must be evaluated on the selected traffic scenes, and confidence thresholds cannot be copied from an unrelated benchmark.

The practical relevance is latency and a simple inference path. The limitation is that frame-level boxes do not provide stable counts by themselves. Repeatedly counting the same vehicle, missed detections under occlusion, and perspective-dependent apparent size require tracking and region rules. Manzil therefore treats the detector as one stage in a measurement pipeline.

3.2.2 Multi-Object Association

Wojke et al. extended a simple online tracker with a learned appearance metric, improving association through longer occlusions [6]. The method illustrates a tradeoff: appearance information can reduce identity switches but adds a feature-extraction model and domain sensitivity. It is a useful alternative if traffic identities fragment severely, but its additional cost must be justified by the metrics that Manzil actually needs.

ByteTrack instead associates high-confidence detections first and then attempts to recover objects from lower-confidence detections rather than discarding them [7]. The authors evaluated the approach across multiple tracking benchmarks and reported strong MOT17 accuracy with real-time throughput on their specified hardware. This is relevant because traffic occlusion often lowers confidence. Its benchmark speed cannot be adopted as Manzil's performance target, and an identity-preserving tracker does not by itself determine lane membership or queue state.

Together, these studies motivate an empirical tracker selection. The decision should be based on count error, identity stability where needed, and end-to-end throughput on the team's hardware. If aggregate counts are reliable without expensive appearance features, the simpler configuration may be preferable.

3.2.3 Traffic Datasets and Domain Coverage

BDD100K provides a large and diverse driving dataset with multiple tasks, including detection and tracking, collected across varied locations, weather, and times [8]. This diversity is useful for initializing or stress-testing a perception pipeline. The camera perspective is primarily vehicle-mounted, however, whereas Manzil may use roadside or elevated recorded video. The dataset cannot substitute for validation on the intended viewpoint.

UA-DETRAC is a benchmark focused on vehicle detection and tracking in traffic-surveillance video [9]. Its fixed-camera scenes are closer to an intersection-observation use case and its protocol makes detector-tracker interactions visible. Its scenes still do not guarantee representation of Lahore traffic composition, motorcycle density, rickshaws, lane discipline, or camera mounting. The strongest plan is therefore to use licensed public material for reproducible comparison and a small, separately documented local sample for domain analysis if collection is feasible and ethically permitted.

These dataset studies also shape data splitting. Adjacent video frames are highly correlated. Random frame splitting can leak nearly identical scenes into training and test sets. Manzil will split by sequence or source and will retain a held-out set that is not used for threshold tuning.

3.3 Conventional and Adaptive Signal Control

3.3.1 Fixed-Time Baseline

A fixed-time controller remains necessary even when it is not expected to dominate every scenario. It supplies a stable reference, exposes whether demand generation itself is biased, and enables exact reproduction. The plan will specify phases, yellow and all-red transitions, cycle length, offsets if any, and demand assumptions. A weak or arbitrarily misconfigured fixed baseline would exaggerate improvements and undermine the comparison.

3.3.2 Max-Pressure Control

Varaiya formalized max-pressure control for networks of signalized intersections [10]. Pressure is derived from upstream and downstream queue conditions, and selecting phases that reduce network pressure has a stronger theoretical basis than using an unexplained weighted score. The method is decentralized and responsive, which makes it a valuable non-learning reference.

Its assumptions and measurements still matter. Queue observations must be available, movements and saturation behavior must be represented correctly, and phase switching must respect minimum green and clearance. Manzil's first rule-based controller will be deliberately transparent; a max-pressure-inspired score is a stronger candidate than ad hoc percentages if the required downstream queues are reliable. Fairness and maximum-red safeguards remain explicit even when pressure is used.

3.4 Reinforcement Learning for Traffic Signals

3.4.1 Cycle-Level Deep Q-Learning

Liang et al. used deep reinforcement learning to adjust traffic-light cycle behavior from traffic state [11]. The study demonstrates how state observations and action granularity can be connected to a learned controller. It also illustrates a recurring limitation: a policy evaluated in a particular intersection and demand setting may not generalize to unseen networks or demand distributions.

For Manzil, deep Q-learning is feasible only after the action space has been bounded and a stable baseline exists. The report does not commit to reproducing the paper's network or treating its reported improvement as a project expectation. The applicable lesson is to define decision interval, phase legality, and reward terms before training.

3.4.2 Pressure-Informed Reward Design

PressLight connects an RL reward and state representation to max-pressure traffic theory [12]. The paper argues that a transportation-grounded reward reduces dependence on arbitrary handcrafted objectives and evaluates the method against conventional and learning baselines in arterial settings. This is directly relevant to Manzil because a reward based only on total waiting time can obscure spillback or starvation.

PressLight does not remove the need for careful evaluation. Network topology, flows, phase definitions, and training budget influence performance. Manzil can use pressure as a principal reward signal or baseline concept, but it must report the exact state and movement mapping. Emergency priority will be implemented as a safety-constrained supervisory mode rather than silently folded into normal-controller reward during the first comparison.

3.4.3 Phase Competition and Network Cooperation

FRAP represents competition among phases and was designed to exploit invariance across phase configurations [13]. This is valuable where equivalent movements should not require independent learning from scratch. Its structured representation increases implementation complexity and still depends on an accurate phase graph. Manzil may adopt its representational lesson without claiming to reproduce FRAP.

CoLight uses graph-attention mechanisms to support cooperation among intersections [14]. The study addresses the limitation of isolated agents by allowing neighboring traffic information to influence decisions. Such coordination is relevant for a multi-intersection corridor, yet attention-based multi-agent training can exceed an undergraduate project's tuning budget and complicate attribution. The initial RL scope should therefore use either a centralized policy for a small network or a clearly defined independent-agent arrangement, with coordination as a measured design choice.

Chu et al. studied multi-agent deep RL for larger signal networks and emphasized scalable coordination [15]. Like CoLight, the work supports the importance of communication and local observations. It also reinforces the risk of non-stationarity: when multiple agents update policies, each agent experiences an environment affected by changing peers. Manzil should not claim network-level RL simply because multiple signals exist; the policy ownership and training scheme must be explicit.

3.4.4 Simulation Environments

CityFlow was introduced as a multi-agent RL environment for large-scale city traffic scenarios [16]. Its contribution is chiefly simulation throughput and support for large networks. SUMO is retained for Manzil because the proposal already depends on its microscopic model, traffic-light control, map

toolchain, and TraCI integration [1]. CityFlow remains an important alternative in the design tradeoff: it may train faster in some large-scale settings, but switching simulators would add model-equivalence work and reduce alignment with the approved proposal.

Lopez et al. describe SUMO's architecture and microscopic simulation capabilities [1]. Simulation provides repeatability and access to internal state, but a simulator is not a validation of real-world impact. Car-following parameters, route demand, signal programs, and map cleanup can materially affect results. Manzil will version those artifacts and frame its conclusions as performance within declared simulation conditions.

3.5 Dynamic Routing

Dijkstra's shortest-path method finds minimum-cost paths on a graph with nonnegative edge weights [17]. Hart, Nilsson, and Raphael introduced the formal basis of A*, which adds a heuristic estimate to guide the search [3]. These algorithms are well established; using them is not Manzil's novelty. The project-specific work is the definition and update of valid edge costs, the treatment of closures, and the interaction between routing and signal priority.

For time-dependent routing, physical distance is insufficient. Manzil proposes expected link travel time plus non-overlapping estimates of queue discharge and residual signal delay. Each term must be expressed in seconds, or normalized with a documented interpretation. If travel time already includes observed queue delay, adding a second congestion penalty would double-count the same effect. Dijkstra is the reference implementation. A* may be used when a lower-bound free-flow travel-time heuristic remains admissible under nonnegative added delays.

Dynamic rerouting can itself be harmful if small fluctuations repeatedly switch the route. The selected path will be changed only when the remaining cost improvement exceeds a threshold and persists for a defined interval, or when the current route becomes infeasible. Route revisions and computation time are part of evaluation, not merely logs.

3.6 Emergency Routing and Signal Preemption

Humagain et al. systematically reviewed emergency-vehicle route optimization and preemption methods [18]. The review identifies recurring gaps: limited real-world application, insufficient use of dynamic traffic data, weak treatment of travel time, effects on general traffic, and safety. This supports Manzil's integrated evaluation, but it does not justify a claim that no integrated systems exist. It also motivates measuring ordinary-traffic disruption alongside emergency benefit.

Qin and Khan studied transitions into emergency preemption and recovery to normal timing [19]. Their work explicitly addresses safe clearance, use of otherwise unnecessary green time, and recovery queues. The key design lesson is that preemption is a lifecycle rather than a single "set green" command. Manzil therefore models requested, safe-transition, active, passage, and recovery states, and retains timing values as configurable simulation policies rather than certified field parameters.

Louati et al. investigated multi-agent, queue-aware emergency crossing at signalized intersections [20]. Queue-aware prioritization is relevant because a green indication alone does not guarantee an unobstructed approach. The method is more ambitious than Manzil's core single-ambulance case, but it motivates look-ahead activation and checking whether downstream storage is available. Manzil will not infer safety merely from the absence of simulated collisions.

3.7 Synthesis and Research Gap

The literature establishes reliable building blocks, but also exposes integration risks. One-stage detection is fast enough to investigate traffic video, yet traffic indicators require tracking, geometry, and local validation. Max-pressure and RL research offers responsive control, yet comparison is sensitive to state, reward, phase constraints, and simulator assumptions. Shortest-path algorithms are mature, yet their usefulness depends on dynamic costs and route-stability rules. Emergency-preemption research stresses transition and recovery, yet ordinary-traffic impact is often a central tradeoff.

Manzil's proposed contribution is therefore not a new detector, shortest-path algorithm, or RL family. It is the design and controlled evaluation of an integrated, traceable prototype in which: (1) perception and simulator evidence are not conflated; (2) normal controllers are compared under matched scenarios; (3) routing and corridor contributions are ablated; and (4) ambulance benefit is reported together with network disruption and recovery. The unresolved research question is whether this integration produces consistent benefit under the selected scenarios and constraints.

Table 3.1: Literature review summary

Study	Method/setting	Reported result in context	Limitations for Manzil	Design implication
Redmon et al. [5]	Unified one-stage object detector on standard image benchmarks	Demonstrated real-time detection and strong speed/accuracy tradeoff	Original benchmark and classes do not establish roadside traffic accuracy	Evaluate a pretrained descendant on selected traffic video
Wojke et al. [6]	Online tracking with learned appearance association	Improved association through occlusion in pedestrian tracking setting	Added compute and domain-dependent appearance model	Consider when identity fragmentation affects counts
Zhang et al. [7]	Two-stage association including low-score boxes	Strong results on several MOT benchmarks; real-time operation on stated hardware	Benchmark speed and identities do not imply calibrated traffic measures	Candidate tracker; measure end-to-end count error
Yu et al. [8]	Large, multi-task driving-video dataset	Provides varied conditions and annotations	Mostly vehicle-mounted viewpoint; domain shift to roadside Lahore scenes	Use for pretraining/stress tests, not sole validation
Wen et al. [9]	Vehicle detection/tracking benchmark for surveillance scenes	Provides joint detection/tracking evaluation protocol	Limited local vehicle mix and site geometry	Prefer sequence-level split and report local limitations
Varaiya [10]	Max-pressure network signal control	Establishes theoretical network-control properties under model assumptions	Requires movement queues and lawful phases	Use as transparent adaptive concept/baseline

Study	Method/setting	Reported result in context	Limitations for Manzil	Design implication
Liang et al. [11]	Deep Q-learning for cycle control	Showed adaptive learned control in simulated setting	Transfer across demand and topology remains uncertain	Bound actions and evaluate only on held-out scenarios
Wei et al. [12]	RL with max-pressure-grounded state/reward	Outperformed tested conventional and learning baselines	Results depend on tested networks and flows	Prefer grounded reward over arbitrary weighted sum
Zheng et al. [13]	Phase competition representation	Learns relations among competing phases	More complex phase encoding	Preserve phase legality and invariance where feasible
Wei et al. [14]	Graph-attention cooperation across signals	Demonstrated network-level cooperation in tested simulations	Training and attribution complexity	Keep first network small; declare coordination model
Chu et al. [15]	Multi-agent deep RL for larger networks	Studied scalable decentralized coordination	Multi-agent non-stationarity and tuning cost	Avoid vague “multi-agent” claim; record ownership
Ma et al. [16]	High-throughput city-scale RL simulator	Supports large traffic-RL experiments	Switching simulator complicates proposal alignment	Retain SUMO; note CityFlow as alternative

Study	Method/setting	Reported result in context	Limitations for Manzil	Design implication
Lopez et al. [1]	Microscopic SUMO simulator architecture	Supports detailed and externally controlled experiments	Simulation parameters determine external validity	Version network, demand, and car-following settings
Dijkstra [17]	Nonnegative shortest-path search	Guarantees minimum path cost under assumptions	Static snapshot unless costs are refreshed	Reference route implementation
Hart et al. [3]	Heuristic minimum-cost path search	Guides search efficiently with suitable heuristic	Invalid heuristic can lose optimality	Use free-flow time lower bound or retain Dijkstra
Humagain et al. [18]	Systematic review of emergency route/preemption work	Identified real-time, safety, deployment, and side-effect gaps	Review does not validate a particular implementation	Measure both EV benefit and ordinary-traffic impact
Qin and Khan [19]	Simulation of entry and recovery strategies	Reported improved transitions over tested conventional approaches	Parameters are not certified for Manzil's network	Implement explicit safe transition and recovery states
Louati et al. [20]	Queue-aware multi-agent emergency crossing	Coordinates emergency passage in tested interrupted intersections	Multi-EV/multi-agent scope exceeds core case	Use queue awareness and downstream checks

3.8 Conclusion

The review supports a conservative technical strategy: start with transparent baselines, define measurements before optimization, select one feasible RL formulation, and model emergency priority as a stateful safety-constrained intervention. It also establishes the need for source-separated video evaluation, matched simulation seeds, ablation of routing and corridor effects, and explicit external-validity limits.

Chapter 4 Software Requirement Specifications

This chapter specifies the behavior required of the Manzil research prototype. Requirements are testable where current evidence permits; numerical values that require benchmarking are explicitly identified as design targets. The specification avoids account management, enforcement, and other modules not justified by the approved scope.

4.1 System Features

The prototype is divided into six feature groups:

1. **Scenario management**: select a network, demand profile, controller, seed, incident, duration, and emergency mission.
2. **Traffic perception**: process licensed recorded video into detections, tracks, lane-region observations, and aggregate measures with timestamps and confidence.
3. **Simulation and normal control**: step SUMO through a single owner, read traffic state, apply fixed or adaptive actions, and preserve legal phase transitions.
4. **Emergency routing and corridor**: validate origin and destination, compute dynamic paths, apply stable rerouting, activate moving priority, detect passage, and recover.
5. **Monitoring**: show map state, signals, traffic indicators, ambulance route, corridor state, warnings, and data-source mode.
6. **Experiment records**: persist immutable configuration, run events, aggregate metrics, software versions, seed, and termination state for comparison and export.

4.2 Functional Requirements

Table 4.1: Functional requirements

ID	Requirement
FR-01	The system shall validate a scenario configuration before starting and shall reject missing network, controller, seed, duration, or demand inputs.
FR-02	The system shall start, pause, resume, and stop one controlled SUMO run while preserving an unambiguous run state.

ID	Requirement
FR-03	The simulation adapter shall be the sole owner of SUMO stepping and signal actuation.
FR-04	The perception worker shall process a selected recorded video and emit timestamped detections, tracks, source metadata, and processing status.
FR-05	The observation service shall aggregate valid vehicle observations into defined road, lane, or approach intervals without silently mixing sources.
FR-06	The operator shall be able to select fixed-time, rule-adaptive, or the configured RL controller before a run.
FR-07	Every signal action shall identify its intersection, requested and applied phase, effective simulation step, controller, and reason.
FR-08	The system shall enforce minimum green and configured yellow/all-red clearance before incompatible movements receive right of way.
FR-09	The emergency service shall validate ambulance origin and destination and return a clear no-route response if either is invalid or disconnected.
FR-10	The route service shall compute a valid remaining path using nonnegative time-based edge costs and shall record the cost snapshot.
FR-11	The rerouting service shall re-evaluate the remaining path after a closure, invalidated edge, or sustained cost improvement above the configured threshold.
FR-12	The corridor coordinator shall activate priority only for a bounded look-ahead set of intersections on the current ambulance route.
FR-13	The corridor coordinator shall detect passage or cancellation and restore the affected intersection to a safe normal-control state.
FR-14	A route revision shall cancel priority requests that no longer belong to the remaining path before activating new ones.
FR-15	The dashboard shall display the current source mode, simulation status, controller, signal state, traffic indicators, ambulance position, route, and corridor state.
FR-16	The dashboard shall present stale, invalid, disconnected, and fallback states visibly rather than continuing with apparently current data.
FR-17	The system shall record configuration, seed, component versions, periodic metrics, route revisions, signal events, failures, and termination reason for each run.

ID	Requirement
FR-18	The comparison service shall align selected runs by metric definition and warn when network, demand, duration, warm-up, or measurement versions are incompatible.
FR-19	The operator shall be able to export run configuration and metrics in a documented machine-readable format.
FR-20	On loss of the learned controller or invalid action, the control service shall fall back to a configured safe baseline and record the transition.

4.3 Quality Attributes and Non-Functional Requirements

Table 4.2: Non-functional requirements and design targets

ID	Attribute	Requirement or target
NFR-01	Reproducibility	A run shall store network and demand identifiers, seed, controller configuration, software/model versions, warm-up, duration, and metric schema.
NFR-02	Safety in simulation	Incompatible green movements shall never be commanded intentionally; all emergency transitions shall use the configured clearance sequence.
NFR-03	Auditability	Applied decisions shall be distinguishable from requests, and their simulation timestamps and reasons shall be retained.
NFR-04	Freshness	Each observation shall carry produced-at and effective simulation time. Stale thresholds are configuration values and stale data shall not silently drive control.
NFR-05	Responsiveness	The control action for step t should be available before its configured decision deadline. The exact deadline is a design target to be set after baseline profiling, not an achieved value.
NFR-06	Isolation	Background video inference and dashboard clients shall not advance SUMO or directly actuate signals.

ID	Attribute	Requirement or target
NFR-07	Usability	A trained operator should be able to configure the core demonstration from one experiment screen, with units and validation messages visible.
NFR-08	Portability	Services should be runnable from versioned environments; containerization is proposed after local baseline integration is stable.
NFR-09	Maintainability	Component interfaces shall use typed, versioned data contracts and controller implementations shall share one interface.
NFR-10	Data integrity	A completed run configuration shall be immutable; derived summaries shall retain their metric version and source run identifiers.
NFR-11	Privacy	Video shall be limited to licensed or team-authorized material, stored only as needed, and excluded from exports unless explicitly requested.
NFR-12	Availability	The prototype shall fail visibly and recover to a safe stopped or baseline-controlled simulation state; no public-service availability is claimed.

4.4 Assumptions and Dependencies

SUMO and TraCI are required for controlled traffic experiments [1, 2]. A selected network has legal lane connections and signal plans after import and cleanup. Video sources have sufficient quality and permission for the selected evaluation. The simulation uses one clock owner. Edge costs are nonnegative. Emergency origin and destination lie on ambulance-accessible edges. The dashboard is an operator tool, not an autonomous public dispatch system.

React and FastAPI are fixed by the approved proposal for the interface and application service boundary [21, 22]. Python is implied by the planned perception, RL, and TraCI ecosystem. PostgreSQL, a WebSocket transport, Docker, and specific chart or map libraries are proposed dependencies and remain replaceable until implementation confirms them.

4.5 Hardware and Software Requirements

Table 4.3: Development and demonstration environment

Area	Development/training need	Demonstration need
Compute	Multi-core CPU; CUDA-capable GPU is desirable for detector/RL experiments but exact memory is unconfirmed	CPU able to run selected SUMO network and services; GPU only if live video inference is demonstrated
Memory/storage	Sufficient for environments, video, checkpoints, run logs, and repeated traces; capacity to be measured	Selected scenario, service builds, model, and demonstration records
Core software	Python, SUMO/TraCI, detector/tracker framework, RL library, Node.js/React toolchain, Git	Version-pinned runtime and browser
Data tools	OpenStreetMap import, SUMO NetEdit, annotation tooling	Prepared network and approved demonstration clips
Persistence	File-backed prototype acceptable initially; PostgreSQL proposed for structured records	Same schema/version as evaluation records

4.6 Use Cases

The primary actor is a research operator. SUMO, the video source, and the experiment store are supporting systems. Evaluators are read-only consumers of dashboards and exported records. Figure 4.1 presents the coverage; Tables 4.4–4.11 define the principal flows.

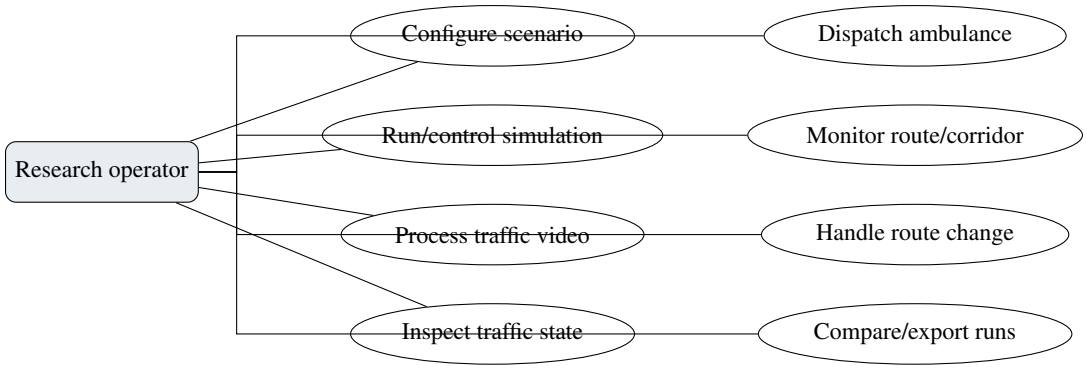


Figure 4.1: Proposed operator use-case diagram

Table 4.4: UC-01 Configure a scenario

Name / actor	UC-01 Configure a scenario / Research operator
Preconditions	Network and controller descriptors are available.
Trigger	Operator requests a new experiment.
Main success flow	Select network, demand, controller, seed, warm-up, duration, source mode, incident, and optional mission; validate; store a draft configuration.
Alternative/failure flow	Reject incompatible controller/network, invalid duration, missing seed, or inaccessible input with field-specific messages.
Postconditions	A valid, versioned configuration is ready; no run has started.

Table 4.5: UC-02 Run and control a simulation

Name / actor	UC-02 Run and control a simulation / Research operator
Preconditions	UC-01 is valid; SUMO adapter is available.
Trigger	Operator selects Start.
Main success flow	Create run record; launch SUMO; complete warm-up; step the simulation; collect state; request and apply lawful actions; publish snapshots; stop at terminal condition.
Alternative/failure flow	If launch/TraCI fails, mark failed and retain logs. Pause/resume does not create competing clock owners. Invalid controller action invokes fallback.
Postconditions	Run is completed, stopped, or failed with an explicit reason.

Table 4.6: UC-03 Process traffic video

Name / actor	UC-03 Process traffic video / Research operator
Preconditions	Licensed video and model configuration are available.
Trigger	Operator selects a video job.
Main success flow	Validate source; decode frames; detect vehicles; associate tracks; map valid tracks to regions; aggregate interval observations; store metrics and job status.
Alternative/failure flow	Unreadable media, unsupported format, model failure, missing calibration, or low-confidence observation is reported; physical queue length is omitted when calibration is unavailable.
Postconditions	Completed observations identify source, model, interval, units, and confidence.

Table 4.7: UC-04 Inspect traffic state

Name / actor	UC-04 Inspect traffic state / Research operator
Preconditions	A run or perception job has emitted data.
Trigger	Operator opens overview or intersection detail.
Main success flow	Display source mode, freshness, counts, queues/proxies, speeds, phases, controller, and warnings; allow time/intersection selection.
Alternative/failure flow	Disconnected or stale streams remain visibly marked; last value is not presented as current.
Postconditions	No system decision is changed; the inspected state is traceable to its source.

Table 4.8: UC-05 Dispatch an ambulance

Name / actor	UC-05 Dispatch an ambulance / Research operator
Preconditions	A running emergency-enabled scenario exists.
Trigger	Operator submits origin and destination.
Main success flow	Validate accessible edges; create mission; compute route/cost; assign simulated ambulance; publish route; request bounded corridor look-ahead.
Alternative/failure flow	Invalid edge or no route returns a reason without partial priority. Duplicate active mission is rejected in the core single-ambulance mode.
Postconditions	Mission is active with a valid current route, or rejected cleanly.

Table 4.9: UC-06 Monitor route and corridor

Name / actor	UC-06 Monitor route and corridor / Research operator
Preconditions	An emergency mission is active.
Trigger	Ambulance advances through the network.
Main success flow	Update position and remaining path; request safe transition ahead; display corridor states; detect passage; release and recover each intersection.
Alternative/failure flow	Missing vehicle, stale position, unsafe request, or corridor timeout triggers cancellation/fallback and a visible alert.
Postconditions	Mission progress and all priority transitions are recorded.

Table 4.10: UC-07 Handle blockage or changed route

Name / actor	UC-07 Handle blockage or changed route / Research operator
Preconditions	An active mission has a feasible current position.
Trigger	Closure or sustained cost change triggers evaluation.
Main success flow	Recompute path from a valid remaining edge; compare benefits; if threshold is met, atomically revise route, cancel obsolete requests, and activate new look-ahead.
Alternative/failure flow	If improvement is too small, retain route. If no route exists, cancel upcoming priority and mark mission blocked.
Postconditions	Exactly one route version is current; every revision has cause and cost snapshot.

Table 4.11: UC-08 Compare and export runs

Name / actor	UC-08 Compare and export runs / Research operator
Preconditions	At least two completed or failed run records exist.
Trigger	Operator selects runs and metrics.
Main success flow	Check comparability; align metric definitions; present distributions and configuration differences; export selected records.
Alternative/failure flow	Warn or refuse direct comparison when network, demand, duration, warm-up, or metric version differs. Failed runs are not silently omitted.
Postconditions	Comparison and export identify source run IDs and schema version.

4.7 Proposed Graphical User Interface

The following figures are proposed wireframes, not screenshots of implemented software. Their purpose is to validate information hierarchy and required operator actions without fabricating progress.

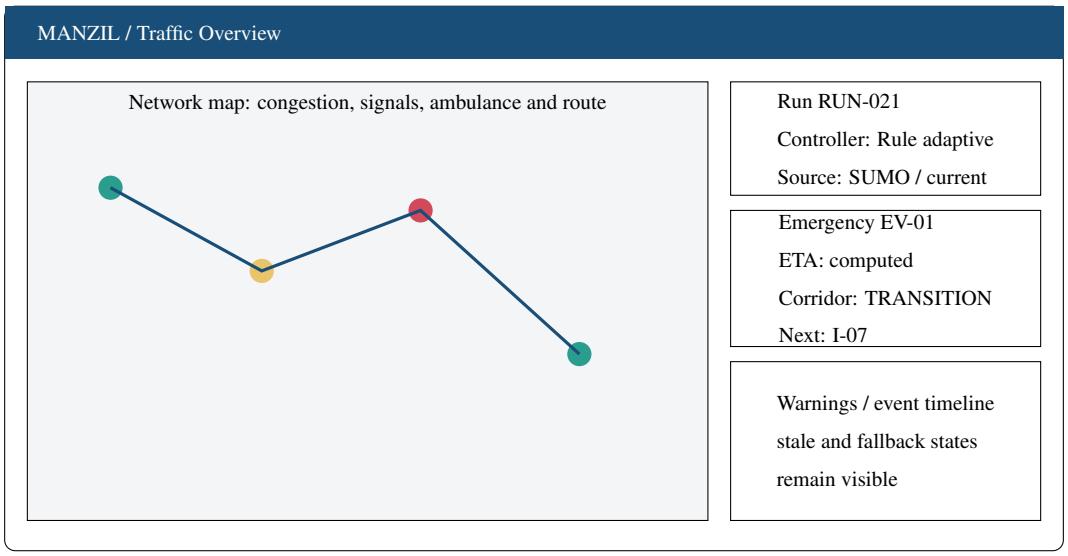


Figure 4.2: Proposed traffic overview wireframe

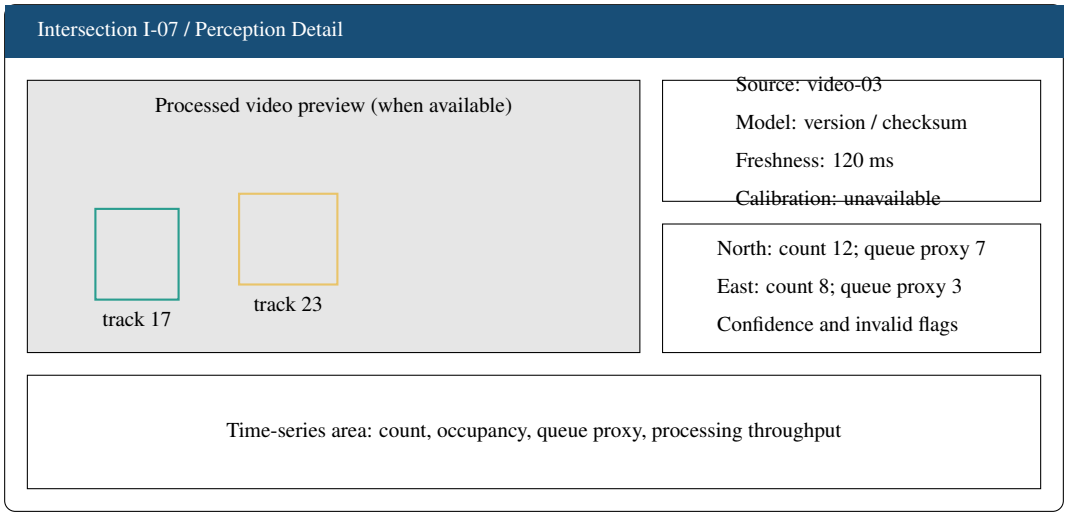


Figure 4.3: Proposed perception and intersection-detail wireframe

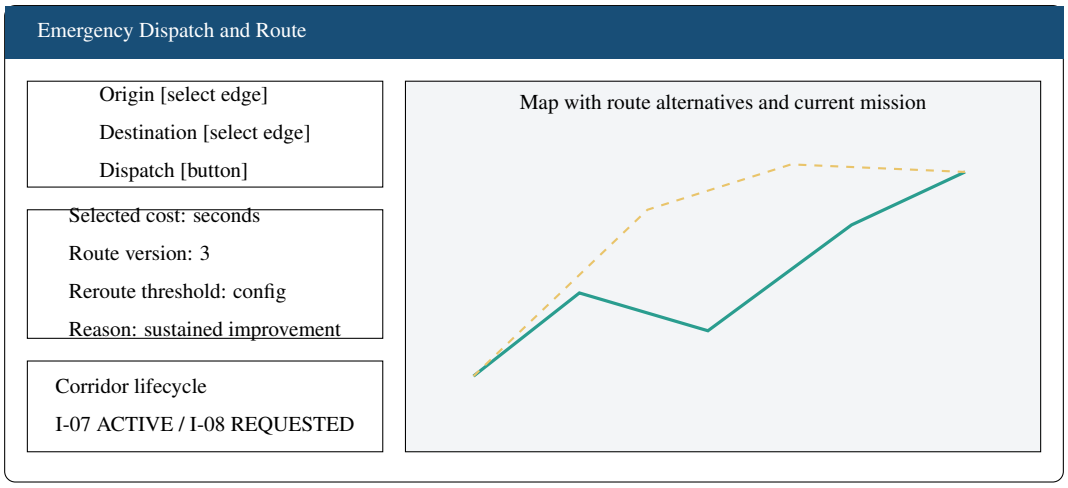


Figure 4.4: Proposed emergency dispatch and route wireframe

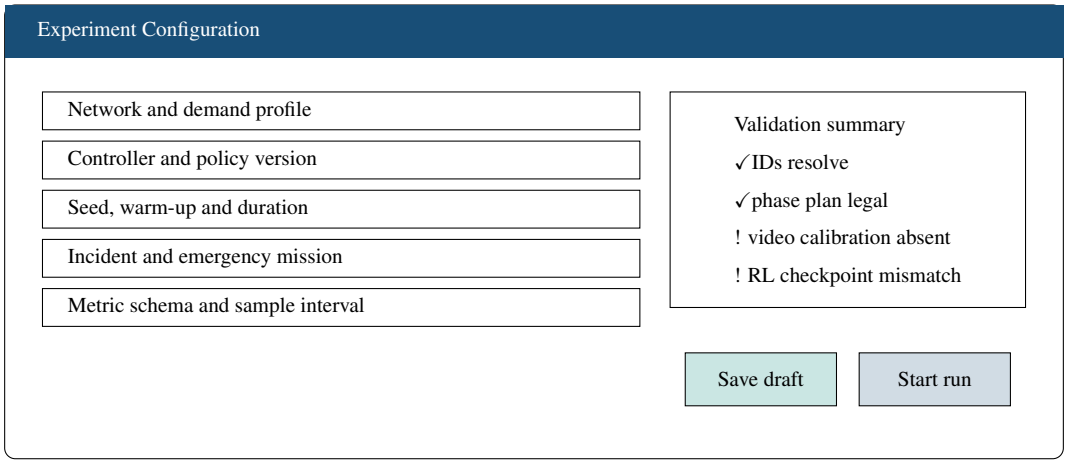


Figure 4.5: Proposed experiment-configuration wireframe

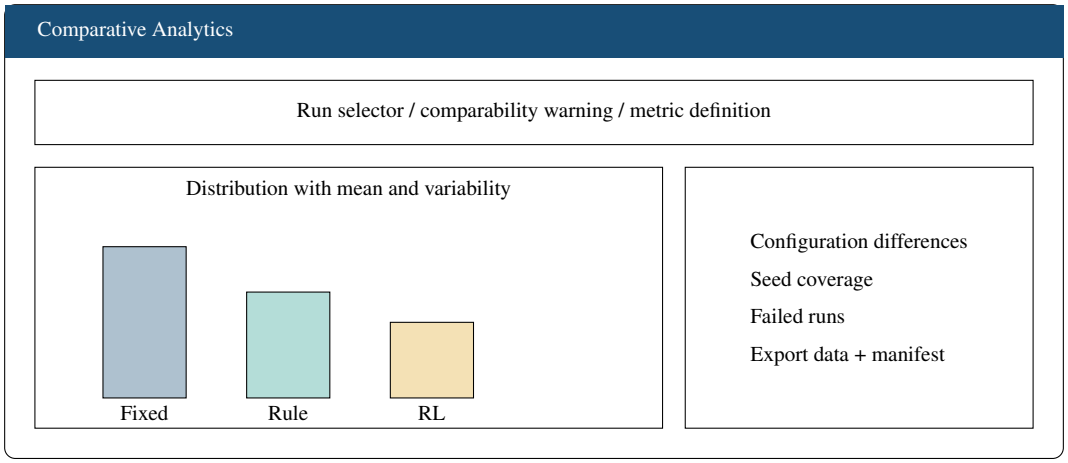


Figure 4.6: Proposed comparative-analytics wireframe; bars are schematic and not results

4.8 Data Design

Persistent data is reserved for configuration, observations, decisions, missions, and outcomes that require audit or comparison. The live road graph, current TraCI subscription, and frame buffers are in-memory runtime state. Raw video remains an external file with a content identifier rather than a database blob.

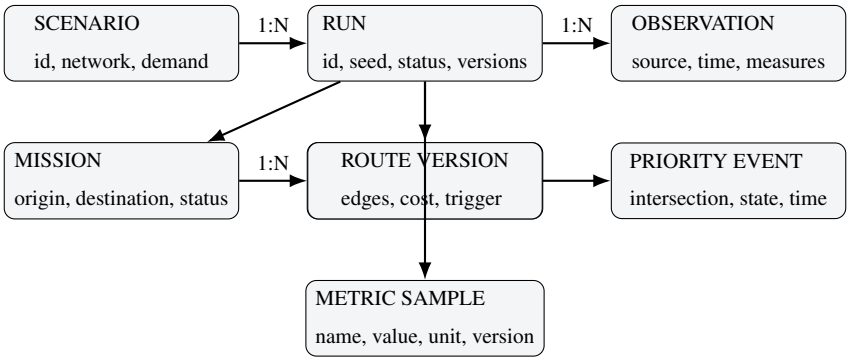


Figure 4.7: Proposed experiment-record entity relationship model

Table 4.12: Data dictionary

Entity	Key fields	Integrity rule
Scenario	scenario_id, network_version, demand_version	References immutable network and demand manifests.
Run	run_id, scenario_id, seed, controller_version, status	Configuration becomes immutable after start; terminal reason required.
Observation	run_id/job_id, source_mode, entity_id, observed_at, valid_until, values, units	Source and units are mandatory; validity interval cannot be inferred.
ControlDecision	run_id, intersection_id, requested_phase, applied_phase, effective_step, reason	Requested and applied actions are separate fields.

Entity	Key fields	Integrity rule
Mission	mission_id, run_id, vehicle_id, origin, destination, status	Only one active core mission per run.
RouteVersion	mission_id, version, edge_list, cost_seconds, trigger, created_at	Version increases monotonically; path begins at valid current position.
PriorityEvent	mission_id, route_version, intersection_id, state, requested/applied time	References the route version that justified priority.
MetricSample	run_id, metric_name, metric_version, value, unit, interval	Comparisons require matching metric name, version, and unit.
Artifact	run/job_id, type, URI, checksum, license/provenance	Content is addressable and provenance is retained.

4.9 Risk Analysis

Table 4.13: Risk register

Risk	Likelihood	Impact	Mitigation	Fallback
Map import lacks valid signals/turns	Medium	High	Inspect and clean a small extract early; validate routes and phases	Use a documented synthetic grid
Local video lacks labels or permission	Medium	High	Use licensed public sequences and record provenance	Evaluate public data only; state domain gap
Occlusion causes unstable counts	High	Medium	Region rules, temporal aggregation, tracker comparison	Limit claims to labelled conditions

RL fails to converge or generalize	Medium	High	Stable baselines, bounded action space, tune only on training scenarios	Report baseline comparison; treat RL as negative/partial finding
Reward improves one metric unfairly	Medium	High	Multi-metric evaluation and starvation constraints	Reject policy violating safety or fairness checks
Rerouting oscillates	Medium	Medium	Benefit threshold, persistence, cooldown, closure exception	Freeze current feasible route
Preemption leaves unsafe phase	Low	Very high	Central state machine and invariant tests	Cancel request and use safe baseline transition
Service timing drifts from simulation	Medium	High	One step owner, simulation timestamps, bounded queues	Run synchronously without live perception
Insufficient compute	Medium	Medium	Profile early; smaller network/model and offline training	Demonstrate recorded inference and compact scenarios
Results are not reproducible	Medium	High	Immutable manifests, seeds, versioned metrics and exports	Exclude untraceable runs

4.10 Requirements Traceability

Table 4.14: Requirements traceability matrix

Requirements	Use cases	Design modules	Planned verification
FR-01, 17–19; NFR-01,10	UC-01, UC-08	Experiment service, store	Schema, validation, replay and comparison tests
FR-02,03,07; NFR-02,06	UC-02	Simulation adapter, control orchestrator	Single-owner and phase-invariant tests
FR-04,05; NFR-04,11	UC-03, UC-04	Perception worker, state aggregator	Annotated sequence evaluation and stale-data tests
FR-06,08,20	UC-02	Controller interface, safety filter	Fixed scenarios, illegal-action and fallback tests
FR-09–11,14	UC-05, UC-07	Graph/routing service	Disconnected graph, closure, hysteresis and path-validity tests
FR-12,13	UC-06, UC-07	Corridor coordinator	Lifecycle, timeout, cancellation and recovery tests
FR-15,16	UC-04, UC-06	API gateway, dashboard	UI state and warning acceptance checks
NFR-03,05,07–09,12	All	Cross-cutting contracts and deployment	Log audit, profiling, usability walkthrough, failure injection

4.11 Conclusion

The specification centers on traceability and failure visibility. It requires one simulation-clock owner, legal signal transitions, stable route revision, explicit source freshness, reproducible run records, and comparisons that cannot hide incompatible configurations. These requirements become procedural and

architectural choices in the next two chapters.

Chapter 5 Proposed Approach and Methodology

This chapter defines how Manzil will be constructed and evaluated. The methodology is designed to answer the research questions with controlled comparisons, not to optimize a demonstration for a favorable single run. Figure 5.1 summarizes the sequence from preparation to evidence.

5.1 Overall Research Workflow

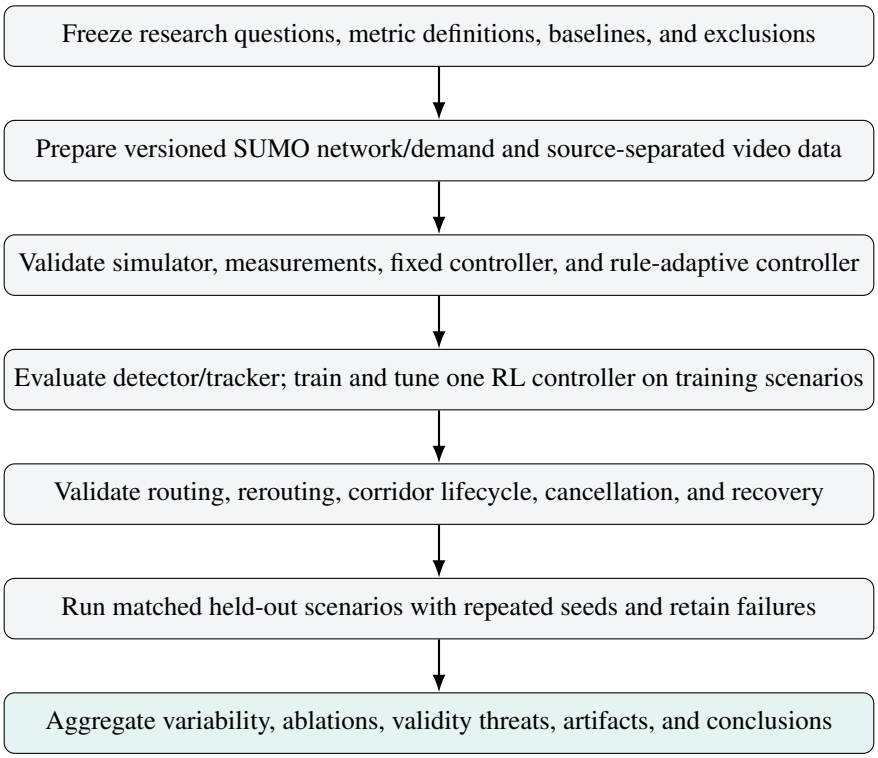


Figure 5.1: Proposed research and evaluation workflow

Two input modes are retained. In *video mode*, annotations supply the reference for perception metrics and no claim is made that video determines the full selected road network. In *simulation mode*, SUMO supplies exact lane, vehicle, signal, and route state for control experiments. A future noise study may inject measured perception error into the simulation-state contract, but it will be included only if the baseline experiments are complete and the protocol is fixed in advance.

5.2 Network and Scenario Preparation

5.2.1 Network Provenance and Validation

A selected, manageable Lahore road area will be exported from OpenStreetMap under its applicable attribution and license conditions [23]. The exact bounding box, extraction date, source file checksum, and conversion command will be stored. SUMO's import toolchain will convert the graph, after which NetEdit inspection will verify lane counts, legal turns, junction types, traffic-light programs, pedestrian-only edges, disconnected components, and ambulance-accessible routes.

The initial size target from project context is approximately 10–20 signalized intersections, but this remains a planning range rather than a confirmed network fact. A smaller validated extract is preferable to a larger inaccurate one. If import quality prevents stable experiments within the schedule, a synthetic multi-intersection grid with documented geometry and demand becomes the fallback; results will be identified as synthetic.

5.2.2 Demand and Incidents

Demand files will define light, medium, and heavy conditions plus an uneven approach case. Vehicle insertion will use explicit random seeds. The warm-up period allows the network to reach a representative occupancy before measurements. Incident scenarios will close an edge, reduce lane capacity, or insert a controlled obstruction at a declared simulation time. A failed insertion due to a full lane will be recorded rather than retried until a favorable run occurs.

The same demand realization will be replayed across controllers where SUMO semantics permit. If a controller changes route choices or insertion feasibility, the manifest will distinguish the original demand request from actual departures. Scenario families will be separated into development, tuning, and held-out evaluation sets.

5.3 Video Perception Pipeline

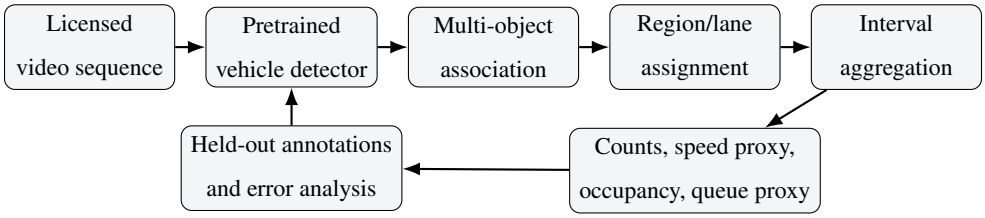


Figure 5.2: Proposed traffic-video perception pipeline

As Figure 5.2 shows, the detector produces class, confidence, and image coordinates; the tracker assigns a temporary identity; and a bottom-center or calibrated road-contact point assigns each track to a polygonal region. Counts use a directed line-crossing or region-entry rule and track history to avoid frame-wise duplication. A speed in physical units requires camera calibration or known scale. Without calibration, the system reports pixel displacement or a categorical slow/stationary state, not kilometres per hour.

For a lane region l over interval k, a simple occupancy proxy is

$$O_{l,k} = \frac{1}{T_k} \sum_{t \in k} \frac{\sum_{i \in V_{l,t}} A(B_{i,t} \cap R_l)}{A(R_l)}, \tag{5.1}$$

where $B_{i,t}$ is a tracked box, R_l is the image region, and T_k is the number of sampled frames. This is image occupancy, not road density. A queue proxy counts stable tracks inside a defined upstream region whose image-motion or calibrated speed is below a threshold for a minimum persistence time.

The detector will be initialized from a documented pretrained checkpoint, consistent with the practical motivation of YOLO-family detection [5]. ByteTrack is the leading proposed tracker because its low-score association may reduce occlusion fragmentation [7]; DeepSORT remains a comparison if appearance association materially improves the selected scenes [6]. The final choice will be based on the held-out count and tracking requirements, not benchmark reputation.

5.4 Normal Traffic Controllers

5.4.1 Fixed-Time Controller

The fixed controller will use a declared cycle and phase sequence appropriate to the selected junction. Green, yellow, and all-red durations and offsets will be versioned. It will not be intentionally starved or given unrealistic timing to make adaptive methods appear superior. A basic saturation check will ensure that at least the light scenario is serviceable.

5.4.2 Rule-Based Adaptive Controller

The initial transparent controller evaluates only legal candidate phases. A pressure-inspired score for phase p is

$$S_p(t) = \sum_{(l,m) \in p} (q_l(t) - \alpha q_m(t)) + \beta w_l(t), \tag{5.2}$$

where q_l and q_m are upstream and downstream queues for a served movement, w_l is the maximum accumulated wait among its approaches, and α, β are documented configuration values. This follows

the motivation of max-pressure control [10] while retaining a direct anti-starvation term.

At each decision instant, the current green is held until minimum green. A phase may be extended if it remains best and maximum green is not reached. A phase whose maximum red exceeds a limit is promoted. Switching inserts the same clearance sequence used by all controllers. No controller can directly bypass the safety filter.

5.4.3 Proposed RL Formulation

One RL formulation will be selected after the rule baseline is stable. A feasible initial choice is a discrete-action value-based policy for a small, fixed phase set; pressure-informed reward design is supported by PressLight [12]. If the network needs coordinated agents and the training budget permits, centralized observations with per-intersection actions are preferred over loosely defined independent learners.

The proposed state for each decision interval includes normalized incoming and outgoing queues, approach waiting time, occupancy, current phase, elapsed phase time, and an emergency-mode flag. An action requests a legal phase or a bounded extension. The safety layer remains authoritative. A candidate reward is

$$r_t = -\lambda_p P_t - \lambda_w W_t - \lambda_s X_t - \lambda_c C_t, \tag{5.3}$$

where P_t is network pressure, W_t normalized waiting, X_t a starvation violation, and C_t an unnecessary-switch penalty. Coefficients and normalization will be fixed on training/tuning scenarios. Emergency performance is evaluated in a separate supervisory experiment so that the first normal-controller comparison is not confounded by rare mission rewards.

Training will record episode seed, demand profile, exploration schedule, optimizer settings, network checksum, reward components, and checkpoints. Evaluation uses deterministic policy settings where applicable and held-out demands. Training return alone is not an evaluation result.

5.5 Dynamic Graph Routing

The routing graph represents legal vehicle movements. Nodes may be junctions or lane/edge transition points; edges store length, allowed class, free-flow time, current travel-time estimate, queue discharge estimate, residual signal delay, closure state, and observation time. The exact representation will match SUMO route validity rather than an abstract map that permits illegal turns.

The primary edge cost is expressed in seconds:

$$c_e(t) = \hat{T}_e^{\text{link}}(t) + \hat{D}_e^{\text{signal}}(t) + \hat{D}_e^{\text{queue}}(t) + M_e(t), \tag{5.4}$$

where $M_e(t) = 0$ for an open edge and infinity for a hard closure. Queue delay is included only if it is not already represented in the link travel-time estimate. Every cost snapshot records source and freshness.

Dijkstra's algorithm provides the reference minimum-cost path under nonnegative weights [17]. A* is permitted with $h(n)$ equal to a lower bound based on straight-line distance divided by the maximum allowed free-flow speed; this remains no greater than true time when added delays are nonnegative [3]. If that condition cannot be guaranteed, Dijkstra is retained.

Data: Current ambulance edge s, destination g, current route R, cost snapshot C_t

Result: Retained or revised valid remaining route

Reject stale or negative costs; apply closures to C_t;

$R' \leftarrow \text{ShortestPath}(s, g, C_t)$;

if R' *does not exist* **then**

    return NO_ROUTE and cancel future corridor requests;

end

Compute remaining costs $J(R)$ and $J(R')$ under the same snapshot;

if *current route is infeasible* **then**

    accept R' immediately;

else

    accept R' only if $J(R) - J(R') > \delta$ for persistence window τ and cooldown has elapsed;

end

Validate route continuity from the current position;

Atomically increment route version, cancel obsolete priority, and publish the revision;

Algorithm 1: Congestion-aware route selection and stable rerouting

The improvement threshold δ, persistence τ, and cooldown are configuration parameters to be tuned before held-out tests. This prevents small state fluctuations from producing route oscillation.

5.6 Moving Green Corridor

The corridor operates on a bounded number of route intersections ahead of the ambulance. Activation may use estimated time to arrival and queue clearance rather than a fixed distance alone. Each intersection receives a request containing mission ID, route version, approach movement, predicted arrival, expiry, and priority level.

Before serving the ambulance movement, the coordinator completes minimum green and applies the configured yellow and all-red clearance for conflicting movements. It does not set every route signal green at once. Passage is confirmed from the simulated vehicle position and lane transition, with a time-out as failure detection. After passage, the intersection enters recovery and either completes a bounded compensation plan or returns at a safe cycle boundary. A revised route cancels obsolete requests. These mechanisms follow the transition and recovery emphasis in emergency-preemption research [19].

Data: Priority request, ambulance state, signal state, safety configuration

if *request is stale, route version is obsolete, or movement is invalid* **then**

    reject and log reason;

end

Queue the request and estimate activation time;

At activation, finish minimum green and apply legal clearance;

Serve the ambulance movement within maximum priority duration;

if *passage is confirmed* **then**

    release and enter recovery;

end

else if *mission is cancelled, route changes, or timeout occurs* **then**

    cancel safely and enter recovery/fallback;

end

Restore normal controller ownership at a valid phase boundary and record recovery time;

Algorithm 2: Corridor lifecycle at one intersection

The core scope contains one active ambulance, which avoids inventing an untested arbitration policy. Multiple emergency vehicles remain future work. Normal-traffic harm is bounded using a maximum priority duration, downstream occupancy check, and maximum-red monitoring. Exact limits are simulation parameters to be justified through sensitivity analysis, not field safety values.

5.7 Integrated Control Loop

Data: Validated run configuration and initialized services

while *run is active and before terminal simulation time* **do**

Advance SUMO by exactly one configured step;

Read subscribed vehicle, lane, signal, and mission state;

Construct a timestamped traffic-state snapshot;

Update route costs and evaluate rerouting at its configured interval;

Update corridor requests and lifecycle states;

if *corridor owns an intersection* **then**

request its safe priority action;

else

request action from the selected normal controller;

end

Validate every requested action through the phase safety filter;

Apply accepted actions through the TraCI adapter only;

Record applied actions, events, and metric samples;

Publish a non-authoritative dashboard snapshot;

end

Finalize run status, artifacts, and checksums;

Algorithm 3: Single-owner simulation and decision loop

Algorithm 3 makes event order explicit. Dashboard refresh and perception jobs cannot step the simulation. A snapshot used at time t carries its effective step so that late data is either mapped deliberately or rejected.

5.8 Experimental Protocol

5.8.1 Comparison Matrix

Table 5.1: Planned experiments and primary evidence

Question	Controlled comparison	Primary measures
RQ1 perception	Detector/tracker output vs. held-out sequence annotations	Precision/recall or mAP where suitable; count MAE; identity metric only when required; queue-proxy error; throughput
RQ2 normal control	Fixed vs. rule-adaptive vs. selected RL under matched scenarios/seeds	Mean travel and waiting time, queue mean/max, throughput, phase changes, max-red/fairness violations
RQ3 routing	Static/reference route vs. dynamic route, holding signal policy fixed	Ambulance travel time, delay, stops, route revisions, computation time
RQ4 corridor	Same routing without vs. with corridor	Ambulance time/stops; ordinary traffic delay/queues; recovery time; safety invariant violations
RQ5 robustness	Demand, incident, seed, and optional observation-noise strata	Effect distributions and sensitivity; failed and no-route runs retained

5.8.2 Scenario Assignment and Repetition

The evaluation manifest will define demand levels, origin/destination pairs, incident timing, warm-up, measurement horizon, and repeated seeds before final runs. Every controller receives the same set. The number of repetitions will be chosen using pilot variance and available computation, then reported rather than retroactively reduced to favorable cases. Averages will be accompanied by standard deviation or confidence intervals, and paired differences will be used when runs share a demand seed.

RL training and hyperparameter selection use only training/tuning scenarios. Held-out demand combinations and at least one unseen seed set are reserved for final evaluation. Perception data is split by source or continuous sequence to reduce temporal leakage. The report will include failures, timeouts, infeasible routes, and safety-filter interventions.

5.8.3 Metric Definitions

For N completed vehicles with travel times T_i, mean travel time is $\bar{T} = N^{-1}\sum_i T_i$. Queue length is sampled at fixed simulation intervals and reported as time mean and maximum by approach. Throughput is the number of vehicles crossing a declared network boundary during the measurement interval. Fairness includes maximum red time and approach-level delay distribution; an aggregate mean alone can conceal starvation.

Ambulance delay is actual travel time minus free-flow route time under the same network definition. Ordinary-traffic impact is evaluated on non-emergency vehicles and includes delay during the mission plus a recovery interval. Route computation time is measured separately from simulated travel time. For perception counts y_k and estimates $\hat{y}_k$, mean absolute error is

$$\mathrm{MAE} = \frac{1}{K}\sum_{k=1}^{K} |y_k - \hat{y}_k|. \tag{5.5}$$

5.8.4 Reproducibility Package

Each run will retain the source revision, dependency manifest, SUMO version, network and route checksums, controller/model version, full configuration, seed, logs, metric schema, and terminal reason. Large external video and pretrained weights may be referenced by license-aware URI and checksum. A script should recreate a run from its manifest; any manual NetEdit change must be committed as a network artifact.

5.9 Threats to Validity

Construct validity is threatened if queue proxies, waiting-time definitions, or reward components do not represent the intended concept. Definitions and units will be versioned, and perception proxies will not be renamed as physical quantities without calibration.

Internal validity is threatened by unmatched demand, different warm-up, hidden seed changes, or controller-specific failures. Paired manifests, one measurement implementation, and retention of failed runs mitigate these risks.

External validity is limited by synthetic demand, map inaccuracies, simulator car-following assumptions, and video domain shift. Findings will be bounded to the tested network, demand families, and video sources. No clinical response-time or public-road safety conclusion will be drawn.

Conclusion validity is threatened by few repetitions, tuning on the test set, and reporting only means.

Pilot variance, held-out scenarios, paired comparisons, variability, and sensitivity analysis will be used. Statistical tests, if used, will name assumptions and effect sizes; significance alone will not establish practical value.

5.10 Conclusion

The methodology specifies a feasible progression from validated baselines to integrated emergency control. It uses time-consistent route costs, hysteresis, explicit corridor transitions, held-out evaluation, and reproducibility artifacts. Most importantly, it defines how negative or mixed results remain valid findings rather than prompting undocumented changes to the experiment.

Chapter 6 High-Level and Low-Level Design

This chapter maps the requirements and methodology to a service-oriented prototype design. The decomposition protects simulation determinism: one adapter owns the simulation clock and all signal actuation, while the API, dashboard, perception, controllers, and experiment store exchange typed messages without competing for control.

6.1 System Overview and Design Goals

The system must support repeatable research runs, interactive demonstrations, and post-run analysis. Its principal design goals are: deterministic ownership of state changes; replaceable controller and perception implementations; explicit time, units, and validity; safe corridor transitions; immutable experimental configuration; and visible failure. Performance is important only after these properties are preserved.

The design assumes a single machine or small development deployment for Deliverable 2/FYP-I. Separating components logically does not require premature distribution. In-process modules may be used initially behind the same interfaces, then isolated as workers if profiling justifies it.

6.2 High-Level Architecture

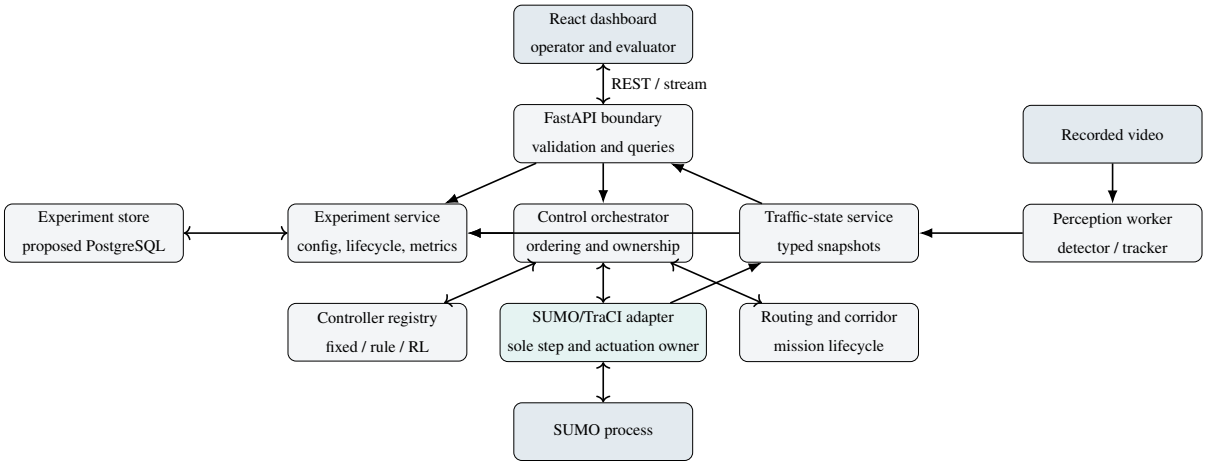


Figure 6.1: Proposed high-level architecture

The dashboard never calls TraCI. It sends validated commands to FastAPI and consumes views. The orchestrator executes Algorithm 3; the adapter alone advances SUMO and applies signals. Controllers return requests, not side effects. The routing and corridor service maintains mission/route versions and intersection priority states. The state service normalizes simulator or perception observations but preserves their source. The experiment service stores configuration and evidence.

6.3 Component Responsibilities and Interfaces

Table 6.1: Component responsibility allocation

Component	Consumes	Produces / owns
React dashboard	Query responses, event snapshots, user input	Presentation state; no control authority
FastAPI boundary	HTTP/stream commands, identity-free local operator session	Validation, command/query boundary, error mapping
Experiment service	Valid configuration, metric/event batches	Run lifecycle, immutable manifest, comparison eligibility
Control orchestrator	Run config and ordered state snapshot	Decision schedule, module ordering, requested actions
SUMO/TraCI adapter	Validated step and action commands	Simulation process, clock, subscriptions, applied actions
Traffic-state service	SUMO observations or perception messages	Versioned source-specific snapshots and freshness
Perception worker	Video and model configuration	Detections, tracks, aggregates, job health
Controller registry	State, legal phases, timing constraints	Requested normal-control action and explanation
Routing service	Graph, mission, cost snapshot	Route versions, no-route reason, reroute decision
Corridor coordinator	Route version, ambulance state, signal state	Priority request/state, release and recovery requests
Experiment store	Validated records and artifacts	Durable configuration, events, metrics, checksums

6.4 Architectural Decisions and Tradeoffs

Table 6.2: Architectural decision record

Decision	**Selected approach**	**Alternative and reason not selected initially**	**Consequence**
Simulation ownership	One orchestrator/TraCI adapter controls stepping and signals	Multiple services calling TraCI risks nondeterministic order	Simple replay and concurrency model
Simulator	SUMO, as approved and supported by detailed road/signal tooling	CityFlow offers training throughput but changes the approved environment	Lower migration risk; performance profiled later
API/UI	FastAPI plus React, fixed by proposal	Server-rendered monolith would simplify deployment but reduce clear service contracts	Typed API and independent UI iteration
Controller integration	Common request interface plus safety filter	Direct controller access to TraCI couples research code to simulation	Comparable controllers; applied action may differ from request
Emergency control	Supervisory corridor state machine	Adding emergency reward directly to normal RL confounds attribution	Clear ablation and safety ownership
Routing	Dijkstra reference; A* only with admissible lower bound	Distance-only route is simpler but ignores the research question	Time-based, auditable costs
Persistence	Structured experiment schema; PostgreSQL proposed	Flat files are quick but weak for cross-run integrity	Engine remains replaceable behind repository interface

Decision	Selected approach	Alternative and reason not selected initially	Consequence
Perception execution	Background worker with bounded messages	In-request inference blocks API and control loop	Adds job state and queue management

6.5 Domain Model and Class Diagram

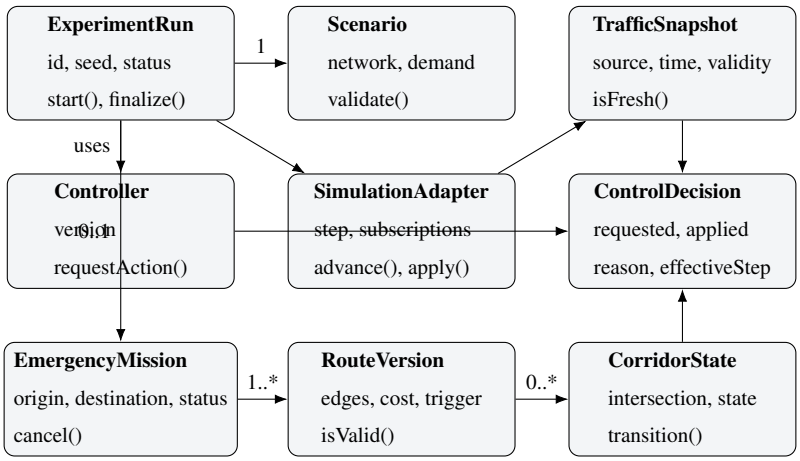


Figure 6.2: Proposed domain model/class diagram

The domain model deliberately separates a requested decision from an applied decision. A controller can request an illegal early switch, but the safety filter and adapter determine the action actually applied. RouteVersion is immutable and corridor states refer to one version, allowing obsolete requests to be cancelled reliably.

6.6 Interaction Sequences

6.6.1 Normal Control Update

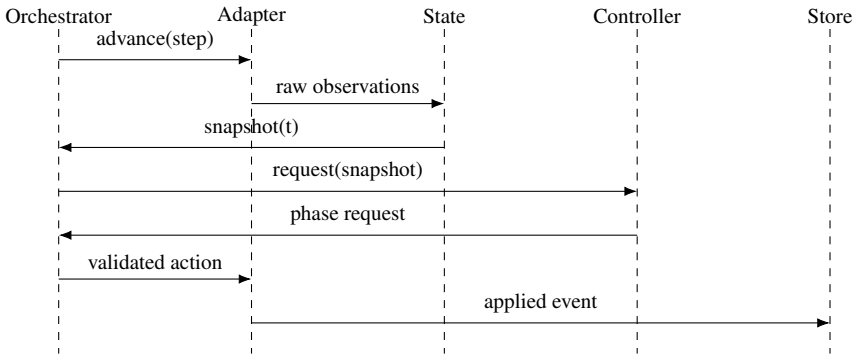


Figure 6.3: Sequence for a normal signal-control update

The snapshot is created only after the simulation step completes. A controller never observes a partial update. If a decision interval spans several simulation steps, the orchestrator reuses the last applied action while metrics continue to sample at their specified interval.

6.6.2 Ambulance Dispatch and Corridor Activation

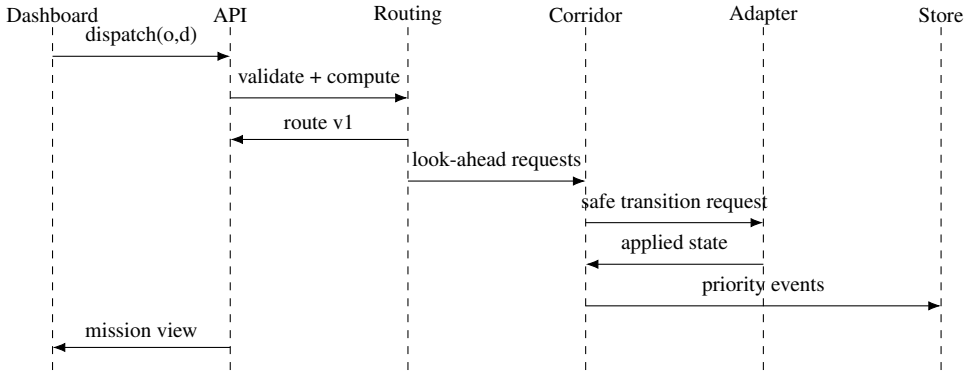


Figure 6.4: Sequence for ambulance dispatch and corridor activation

6.6.3 Rerouting and Recovery

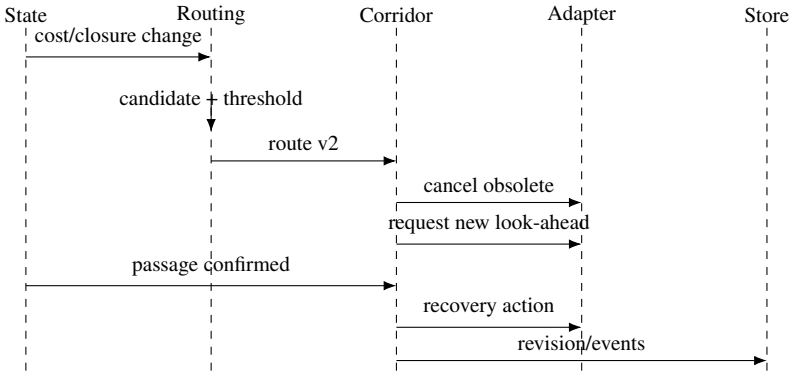


Figure 6.5: Sequence for route revision, obsolete-priority cancellation, and recovery

6.7 Corridor State Model

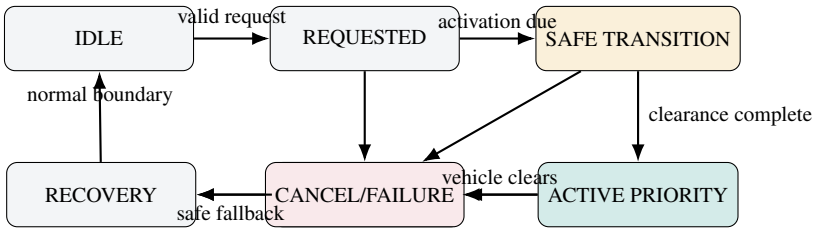


Figure 6.6: Intersection-level moving-corridor state transitions

State transitions are event-driven and guarded. REQUESTED does not imply signal ownership. SAFE TRANSITION waits for the normal phase's minimum service and clearance. ACTIVE PRIORITY has a maximum duration. PASSAGE is a recorded event, and RECOVERY must complete before the normal controller resumes direct ownership. Cancellation never jumps directly to an arbitrary normal phase.

6.8 Traffic-State and Event Contracts

Table 6.3: Core interface and data-contract fields

Contract	Required fields	Rule
TrafficSnapshot	schemaVersion, sourceMode, producedAt, effectiveStep, validUntil, networkId	Consumers reject unknown schema or expired validity; wall time and simulation time are distinct.
ApproachState	intersectionId, approachId, count, queue, wait, speed, occupancy, units, confidence	Missing is not zero; uncalibrated physical quantities are omitted.
SignalState	intersectionId, programId, phaseId, phaseElapsed, legalNext, owner	Owner is NORMAL or CORRIDOR; legal next phases come from validated configuration.
ControlRequest	controllerId, intersectionId, requestedPhase, duration/extension, snapshotStep, reason	Request has no side effect and may be rejected or transformed.
AppliedControl	requestId, appliedPhase, effectiveStep, safetyIntervention, reason	This record is authoritative for analysis.
RoutePlan	missionId, routeVersion, edgeIds, costSeconds, snapshotStep, trigger	Edge order must be continuous from valid current position.
PriorityRequest	missionId, routeVersion, intersectionId, movementId, ETA, expiresAt	Obsolete route version or expired request is rejected.

Contract	Required fields	Rule
MetricSample	runId, metricName, metricVersion, interval, value, unit	Aggregation across mismatched versions is prohibited.

Event ordering is based on simulation step within a run and a monotonic sequence number assigned by the orchestrator. Wall-clock timestamps support diagnostics but do not determine traffic causality. Route revisions and priority events are committed in an order that prevents an old request from being interpreted as current.

6.9 Concurrency, Synchronization, and Resource Ownership

The orchestrator is the only writer to simulation time. TraCI access is serialized behind the adapter. Controller computation receives an immutable snapshot. Perception uses a bounded work queue; if it falls behind, the worker marks observations late rather than blocking control. Dashboard streams are lossy views: a slow client may skip intermediate snapshots but cannot delay decisions.

Run configuration is immutable after start. The current route version is updated atomically with the cancellation of obsolete corridor requests. Database writes are batched where safe, but an applied signal event and mission-state transition must not be acknowledged before durable enqueue. On shutdown, the system stops accepting commands, terminates or detaches SUMO safely, flushes records, and gives the run an explicit terminal state.

6.10 Error Handling and Fallback

Table 6.4: Failure handling strategy

Failure	Immediate behavior	Recorded/recovery behavior
SUMO launch/TraCI disconnect	Stop decisions and do not present live state	Mark run failed; retain command, stderr, step, and manifest
Stale observation	Exclude or use declared simulator fallback	Record staleness and selected fallback source
Invalid controller action	Reject through safety filter	Apply fixed/safe continuation and record intervention
RL service unavailable	Switch at valid boundary to configured baseline	Record fallback interval; exclude from clean RL aggregate or report separately
No route	Do not spawn/activate partial corridor	Mark mission blocked; show cause and graph snapshot ID
Lost ambulance/timeout	Cancel future priority and recover active intersections	Mark mission failed/cancelled and preserve last position
Persistence failure	Buffer within a bounded queue; stop before unbounded evidence loss	Mark run incomplete if durability cannot be restored
Dashboard disconnect	Continue experiment unaffected	Reconnect from current snapshot and event cursor

6.11 Detailed Module Policies

6.11.1 Controller Policy

All controllers implement a common method accepting a snapshot, intersection configuration, and decision context, and returning a request plus explanation fields. They cannot write global state or call TraCI. Deterministic evaluation disables exploration and seeds all permitted stochastic operations. The safety filter verifies phase membership, minimum green, clearance, maximum priority duration, and ownership.

6.11.2 Routing Policy

The graph builder derives edges from validated SUMO movements. Cost providers are composable but must state units and whether link time already includes queue delay. A route plan is immutable. The mission points to one current version, and a revision is accepted only through Algorithm 1. Closures override cost and invalidate affected routes immediately.

6.11.3 Persistence Policy

The store uses migrations and foreign keys for structured data, and checksummed external artifacts for large files. Raw high-frequency state may be downsampled for persistence, but the interval and aggregation method are recorded. Secrets or local filesystem paths are not exported as research data. Database selection will be confirmed by implementation; the domain schema is independent of that choice.

6.11.4 User-Interface Policy

Colors are supplemented with labels and icons so congestion or failures are not conveyed by color alone. Units and source mode appear beside values. Schematic predictions and measured values are visually distinct. The UI does not display “time saved” until a valid reference run exists under the same comparison manifest.

6.12 Testing Strategy

Unit tests cover cost functions, route validity, hysteresis, metric calculations, schema validation, and corridor transition guards. Property or invariant tests generate action sequences to ensure incompatible greens are never produced and that corridor cancellation eventually reaches IDLE through RECOVERY. Integration tests run small SUMO networks to validate step ownership, detector-to-state messages, route application, and persistence. Scenario tests replay fixed seeds for normal control, closure rerouting, no-route, controller failure, stale data, ambulance timeout, and dashboard disconnect.

Research verification is separate from software correctness. Passing unit tests shows that the implementation follows its specification; it does not show that the policy improves traffic. Comparative experiments described in Chapter 5 provide that evidence.

6.13 Coding, Build, and Maintainability Policies

Python modules will use static type hints, formatting and linting, bounded dependency versions, structured logging, and docstrings for public interfaces. React components will separate data fetching from presentation. Configuration is stored as validated files rather than scattered constants. Continuous checks should run unit tests and a minimal headless SUMO scenario. Releases used for experiments are tagged and their commit ID is written into run manifests.

The intended repository structure separates application services, controllers, routing, perception, simulation, dashboard, schemas, scenarios, and experiment artifacts. This is a proposed organization, not evidence that those directories already exist. Small interfaces allow a tracker, RL algorithm, database, or dashboard transport to change without rewriting the simulation loop.

6.14 Conclusion

The design assigns one clear owner to every mutable resource and makes source, time, units, route version, and applied action explicit. The architecture supports the intended progression from baselines to RL and from static routing to a safe moving corridor, while allowing failures and negative experimental outcomes to be recorded honestly. It is sufficiently modular for team implementation without requiring distributed-system complexity at the prototype stage.

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Chapter A Deliverable 2 Requirements Coverage

Table A.1: Requirements coverage matrix

Deliverable requirement	Primary evidence	Report location
Project purpose and vision	Approved proposal and maintained context	Chapters 1–2
At least 15 relevant studies	Verified peer-reviewed literature and official papers	Chapter 3 and bibliography
Software requirements and use cases	Proposal scope and planned operator workflow	Chapter 4
GUI and data design	Proposed dashboard and persistence needs	Sections 4.7–4.8
Research methodology	Proposal baselines and reproducible evaluation plan	Chapter 5
High- and low-level design	Service responsibilities and control safety constraints	Chapter 6
Architecture, UML, sequences, state model	Original TikZ diagrams	Sections 6.2–6.7
Honest project status	Proposal stage statement and repository evidence	Throughout; Section 1.3

Chapter B Open Items for Team Confirmation

The report remains usable without resolving the items below, but they must be confirmed before an implementation or final-results report makes corresponding claims.

1. Exact Lahore road area, network size, and imported OpenStreetMap snapshot.
2. Final detector checkpoint, tracker configuration, annotated video sources, and license records.
3. Final RL algorithm and hyperparameters after baseline feasibility tests.
4. Database selection, demonstration hardware, and measured update/latency targets.
5. Actual application screenshots, once the interfaces exist; all current interface figures are labelled as proposed wireframes.
6. Any faculty-issued certificate or approval page not present in the supplied FAST template.